

Population structure and gene flow in a heavily disturbed habitat: implications for the management of the imperilled Red Hills salamander (*Phaeognathus hubrichti*)

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Received: 28 June 2011 / Accepted: 15 March 2012 / Published online: 4 April 2012
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Abstract Estimating levels of gene flow and assessing levels of population connectivity are of critical importance to the field of conservation genetics, especially for imperiled species. Many factors can influence dispersal and therefore gene-flow patterns across a natural landscape. These patterns can be substantially altered by the impacts of habitat modification by humans or natural phenomena. Landscape-genetic studies that address both historical and contemporary influences on gene flow can be critical to demonstrating whether isolated populations with low levels of genetic variation are typical of the species or a result of strong negative effects of such modification. We used 10 microsatellite markers to investigate the spatial genetic patterns of the Red Hills salamander (*Phaeognathus hubrichti*), a federally listed species. Bayesian clustering revealed five well-supported demes within the range of *P. hubrichti*. Gene-flow analysis suggested that overall migration levels for *P. hubrichti* are low, but coalescent methods indicate that migration levels were significantly higher before habitat modification by humans. By accounting for history and species characteristics, our results suggest that loss and fragmentation of habitat have strongly negatively affected *P. hubrichti* by reducing

migration, increasing bottlenecks, and promoting high levels of inbreeding.

Keywords Conservation genetics · Red Hills salamander · Landscape genetics · Gene flow · *Phaeognathus hubrichti*

Introduction

The identification of intrapopulation genetic structure and diversity is fundamental to many fields within evolutionary biology and ecology. For conservation biology this information has become increasingly important in conservation planning for imperiled species (see, e.g., Geist and Kuehn 2005; Pabijan et al. 2005; Dixon et al. 2006; Schwartz et al. 2007; Marshall et al. 2009; Matern et al. 2009; Straub and Doyle 2009). Maintaining genetic diversity, and therefore evolutionary potential, is fundamental to the long-term survival and recovery of at-risk species (Avice 2004; Morgan et al. 2008). Theoretical models have demonstrated that the maintenance of genetic diversity and population viability is critically dependent on gene flow among local populations (Gilpin and Hanski 1991; Harrison 1991; Swindell and Bouzat 2005). This transfer of genes between populations influences a number of evolutionary processes, such as population persistence, release from inbreeding depression, and adaptive response (Frankham et al. 2002). A crucial goal of conservation genetics is, therefore, not only to identify the overall genetic structure, diversity, and connectivity between populations but also to understand the factors that shape such patterns.

The rapidity of habitat fragmentation has made the estimation of migration and gene flow between populations an increasingly important undertaking for conservation

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biologists. Developing effective management and recovery plans requires understanding of how and to what extent migration occurs in a modified landscape (Goldberg and Waits 2010). Accordingly, research has recently focused on identifying landscape features that impede or facilitate gene flow (see, e.g., Murphy et al. 2008; Zellmer and Knowles 2009; Goldberg and Waits 2010; Shirk et al. 2010), but those features are not static through time, and a quickly changing landscape can make interpretation of current patterns difficult (Zellmer and Knowles 2009).

Clearly, fragmentation affects gene flow in a wide variety of species (Epps et al. 2005; Proctor et al. 2005; Coulon et al. 2006; Cushman 2006; Vandergast et al. 2007; Zellmer and Knowles 2009); yet, without knowledge of a species' historical dispersal patterns, assessing the magnitude of the impact is difficult. For example, current patterns in population structuring and gene flow may reflect the influence of landscape at several temporal scales (Zellmer and Knowles 2009; Anderson et al. 2010; Reed et al. 2011). Failing to take into account the importance of temporal variation in migration patterns and rates can obscure or even alter our interpretation of the influence that habitat alteration exerts on genetic differentiation in natural populations.

The range of the Red Hills salamander (*Phaeognathus hubrichti* Highton), federally listed as threatened (IUCN: Endangered), has been severely fragmented by silvicultural and agricultural practices (Dodd 1991; Godwin 2008; Apodaca 2010). Over 80 % of the species' original habitat is estimated to have been transformed or severely affected by such activity (Apodaca 2010). The majority of this habitat loss is the result of clear-cutting, which not only reduces the overall amount of habitat but also greatly increases habitat fragmentation. Assessing the effects of this human-caused fragmentation is difficult, however, because the range of *P. hubrichti* is also naturally fragmented, as a result of the species' reliance on a set of specific geologic formations (Tallahata, Hatchetigbee, and Nanafalia) and ecological conditions, including steep (generally greater than 30°) slopes and mature hardwood canopy (Schwaner and Mount 1970; Jordan and Mount 1975; Dodd 1991) and the typically poor dispersal of amphibians in general (see, e.g., Blaustein et al. 1994; Pabijan et al. 2005; Measey and Tolley 2011).

In the study reported here, we analyzed 10 microsatellite loci from specimens collected across the range of *P. hubrichti*. Our goals were (1) to define the population structure and genetic variation across the range of *P. hubrichti*, (2) to compare estimates of current and long-term (pre-habitat-alteration) patterns of gene flow, (3) to investigate the effect of habitat modification by humans on recent patterns of migration and demographic events, and (4) to establish the role that landscape plays in shaping genetic patterns so as to provide habitat managers with targets for restoration

and protection. In addressing these questions we seek to provide basic, yet crucial, data on an endangered and secretive species that we know surprisingly little about. We aim to provide information necessary for a successful and comprehensive recovery plan for *P. hubrichti*.

Methods

Population sampling

Tissue samples were collected under U.S. Fish and Wildlife Threatened Species Permit TE136961-0. From each individual, we collected a small tail-clip according to standard protocols. Tissue was immediately preserved in 95 % ethanol and transferred to a −80 °C freezer at the University of Alabama Herpetology Collection. Sampling occurred across the entire known range of *P. hubrichti* but was focused on sites that had been surveyed by Dodd (1991). In all, we collected 105 individuals from 21 localities (Fig. 1).

Microsatellite analysis

We used the Qiagen DNeasy tissue kit and protocol (Qiagen, Inc., Valencia, CA) to extract DNA from all tissue samples. We amplified 10 microsatellite markers using polymerase chain reaction (PCR). Each primer was developed specifically for use in *P. hubrichti* (Lance et al. 2009). Information on each primer as well as PCR conditions is given by Lance et al. (2009). Each locus was amplified individually, and labeled PCR products were run on an ABI 3730 Genetic Analyzer. Samples were genotyped with GeneMapper 3.7 software (Applied Biosystems, Inc.). Scoring and quality control of data were done with GeneMarker version 1.7 (Softgenetics, LLC). MICRO-CHECKER version 2.2.3 (Van Oosterhout et al. 2004) was used to check for possible null alleles, linkage disequilibrium, and scoring errors. Quality of data and repeatability were tested by regenotyping of 10 individuals per locus. This test resulted in a repeatability success rate of 98 %. HWE proportions for each population and locus were tested with an approach analogous to Fisher's exact test with a Markov chain of 1,000,000 iterations and 10,000 dememorization steps (Guo and Thompson 1992) in Arlequin version 3.11 (Excoffier et al. 2005).

Population genetic structuring and diversity

We used two Bayesian methods to investigate the genetic structuring of populations. The first was implemented in TESS version 2.3 (Chen et al. 2007). TESS uses hidden Markov random fields to model spatial dependence among individuals (Chen et al. 2007). This approach has the

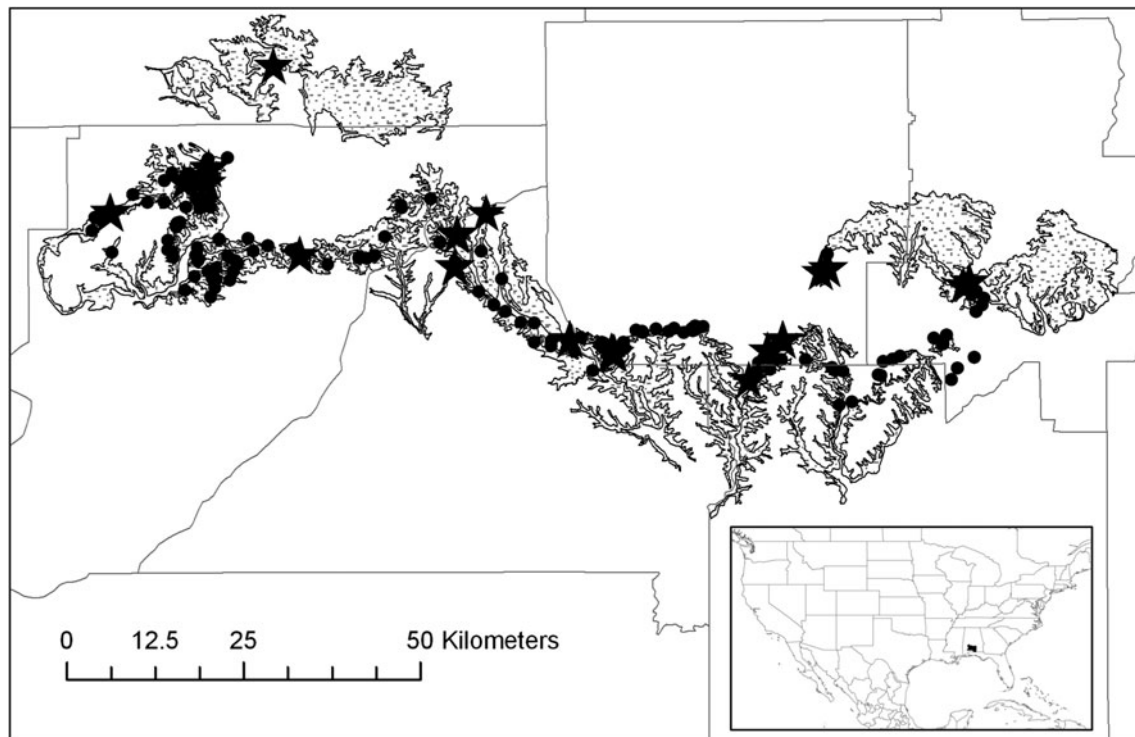


Fig. 1 Range map of *P. hubrichti* with an outline of geologic layers required by the species. Circles are known localities; stars represent tissue-collection sites

advantage that it incorporates the a priori assumption that individuals near one another are more likely to have similar allele frequencies than individuals from widely separated localities. TESS was run for 100,000 simulations with a burn-in of 20,000. To estimate K (where K equals the number of populations), we ran 100 replicates each for K values ranging from 3 to 9. For each K we averaged the 10 best deviance-information-criterion values and plotted them. Once we established the K value, the 10 runs with the lowest deviance-information-criterion values for that K were exported to CLUMPP version 1.1.2 (Jakobsson and Rosenberg 2007). We used the FullSearch algorithm in CLUMPP to match cluster membership over multiple runs. We repeated the above procedure using both the admixture and the no-admixture models implemented in TESS version 2.3 (Chen et al. 2007). Because the models did not differ significantly, we used the results from the no-admixture model, as recommended by the authors. We also studied the spatial genetic patterns by means of STRUCTURE 2.2 (Pritchard et al. 2000). STRUCTURE uses a Bayesian framework to assign individuals to populations by maximizing HWE within populations, and is one of the most commonly used structuring programs. The ΔK method of Evanno et al. (2005) was used to assess the best value of K . For each run of STRUCTURE, the program was run for 1,000,000 MCMC cycles, with a burn-in of 100,000 and default settings. We also used Arlequin version 3.11

(Excoffier et al. 2005) to determine the ability of each Bayesian clustering method to assign the genotypes of individuals to the population from which they came (Paeckau and Strobeck 1995). Log-likelihoods were calculated from the allele frequencies from the observed data and therefore provided the global individual likelihood. The global individual likelihood is the likelihood that the focal individual came from the predetermined population for each locus of the individual's genotype (Cabe et al. 2006).

We investigated patterns of intra- and interpopulation diversity among assigned populations using Arlequin version 3.11. We performed a locus-by-locus analysis of molecular analysis (Excoffier et al. 1992) using 10,000 permutations and computed pairwise and global values of F_{ST} using 1000 permutations for significance followed by a sequential Bonferroni correction. Observed heterozygosity (H_O), expected heterozygosity (H_E ; Nei 1987), allelic diversity, and Wright's (1978) fixation index (F_{IS}) for all populations were also calculated with Arlequin version 3.11. Significance for F_{IS} values was obtained by permutation of haplotypes among individuals within populations 10,000 times in Arlequin version 3.11.

Estimates of short- and long-term gene flow

To estimate the long-term (predisturbance) patterns of gene flow, we used the program MIGRATE version 3.0.3

(Beerli and Felsenstein 1999, 2001). MIGRATE uses coalescent theory and Markov-chain Monte Carlo (MCMC) techniques to estimate long-term (roughly $4N_e$ generation) pairwise gene-flow estimates ($M = m/\mu$, where m = migration rate and μ = mutation rate) and effective population sizes ($\Theta = 4N_e\mu$, where N_e is effective population size). Distributions were estimated in the Bayesian implementation of MIGRATE on the basis of the accuracy of the Bayesian approach under a wider variety of conditions (Beerli 2006). Following the recommendations of the author, we did an initial run on our data using F_{ST} to find the start parameters, and the results of this run were used as start parameters for subsequent runs. Results from different runs were similar, indicating that the Markov chains had probably converged on the stationary distribution. For each run we used the continuous Brownian mutation model and ran 5,000,000 generations per long chain with a burn-in of 100,000. We also used BayesAss+ version 1.3 (Wilson and Rannala 2003) to estimate short-term [within the last few (3–5) generations] migration rates. Although both programs use a Bayesian MCMC approach, BayesAss+ uses a genetic assignment method rather than a coalescent method to estimate gene flow. Genetic-assignment methods tend to estimate short-term dispersal rates, as opposed to coalescent methods, which are closer to long-term averages (Berry et al. 2004; Paetkau et al. 2004). We performed five runs (each with different seed values) of 5 million generations with a 2 million generation burn-in and sampled the chain every 2,000 generations. As with any MCMC approach, both methods used to estimate levels of gene flow carry the assumption that convergence of the MCMC algorithm has occurred. To compare estimates from the two methods, we converted the estimates for M from MIGRATE to proportion of migrants (m) for populations by means of the formula $m = M\mu$, where $\mu = 5.4 \times 10^{-4}$ (Goldstein et al. 1995; Howes et al. 2009).

Short- and long-term gene-flow estimates (m) were compared with a Wilcoxon matched-pairs test in STATISTICA version 6.0. The Wilcoxon matched-pairs test is a nonparametric method requiring the assumption that the samples are not independent of each other (Sokal and Rohlf 2003), so it is ideal for determining whether the two methods differ significantly in migration estimates.

Short-term demographic events

To test for the signature of recent bottleneck events, we used the program BOTTLENECK V1.2.02 (Cornuet and Luikart 1996; Piry et al. 1999). BOTTLENECK is based on the theory that, in a population bottleneck, the allelic diversity will decline more rapidly than the heterozygosity. That is to say, because of the loss of rare alleles, the observed heterozygosity is larger than the heterozygosity

expected on the basis of the number of alleles present if the locus were at mutation-drift equilibrium. We ran the infinite-alleles model (IAM) because it was most likely to fit the large number of interrupted and compound microsatellites in our study (Cornuet and Luikart 1996). Because interrupted and compound microsatellites tend to fit the IAM more closely, they are among the most useful markers for detecting bottlenecks by this method (Cornuet and Luikart 1996). We also tested the sensitivity of the model by varying the parameters, using the two-phase model with varying degrees of stepwise mutation model (SMM; 10–40 % in steps of 10). All tests were run with 10,000 permutations. One-tailed Wilcoxon tests were used to determine the significance of heterozygosity excess or deficiency for each population.

Landscape analyses

To examine the effects of landscape features and the specific habitat requirements of *P. hubrichti* on population structure, we used a series of Mantel tests (Wright 1943; Mantel 1967; Smouse et al. 1986) to test for correlations between geographic distance and genetic distance. As true Euclidian distances are often not the best indicator of genetic distance (Funk et al. 2005; Spear et al. 2005; Storfer et al. 2007; Wang 2009), we also tested alternative distance measures. For example, we tested to determine whether the distance between any two collection points along steep slope was more closely associated with genetic distance than with simple Euclidian distances. In addition, we created two resistance layers in ArcGIS V 9.2 using the spatial-analyst extension. The first distance was calculated with a resistance layer created by assignment of steep slopes (greater than 20°) as the most conducive to migration (a resistance value of 1). Non-steep slopes had twice the resistance value of steep slopes (a resistance value of 2). Thus, the distance traveled completely along a slope is counted as a Euclidian distance. This is a valid assumption because *P. hubrichti* requires steep slopes to burrow and probably cannot migrate long distances without suitable habitat (Schwaner and Mount 1970). The second distance was a combination of slope conductance and assignment of penalties to areas of unsuitable geology. In addition to the parameters of the steep slopes model, areas of unsuitable geology also had twice the resistance value (a resistance value of 2) of suitable geology (a resistance value of 1). *Phaeognathus hubrichti* has been found only within three (Tallahata, Hatchetigbee, and Nanafalia) geologic layers, so these layers are highly likely to be essential to the species. Each of these layers was imported into Circuitscape 2.2 (McRae 2006). Circuitscape uses circuit theory to evaluate a total landscape resistance over multiple paths. The advantage is that, in most biological

systems, a single path (as in least-cost path analysis) is highly unlikely to explain gene flow over many generations (Lee-Yaw et al. 2009). Parameterization was based on transect data (Godwin 2008) and expert opinion. All Mantel tests used F_{ST} genetic distances and were conducted with the Isolation By Distance Web Service version 3.16 (Jensen et al. 2005) for 10,000 permutations.

Results

Genotypic data

For the 10 loci sampled across all populations of *P. hubrichti*, 100 alleles were observed. The median number of alleles per locus was 5.94, standard deviation 0.641. Analysis in MICRO-CHECKER version 2.2.3 did not detect the presence of linkage disequilibrium, scoring errors, or null alleles. A total of 3 loci showed deviations from HWE, but because the deviations were not consistent across all populations (i.e. each loci only deviated from HWE in one population), we did not exclude them from the analyses.

Population genetic structuring and diversity

TESS (Chen et al. 2007) and STRUCTURE (Pritchard et al. 2000) each identified five populations within the range of *P. hubrichti* (Fig. 2). Both identified the same populations, whether or not admixture was used (for

TESS). Given the problematic nature of the terminology surrounding populations and how we perceive them, we will refer to each of the clusters identified by these programs as demes for the rest of the study. The likelihood-assignment test (Paetkau and Strobeck 1995) as performed in Arlequin (Excoffier et al. 2005) showed overwhelming support for each of the demes. Consistent with these results, the analysis of molecular variance (Table 1) indicated a high amount (15.58 %) of variation between populations, as well as a high global fixation index (0.1558). Pairwise F_{ST} values were also high (Table 2); each population pair had a value above 0.13. All F_{ST} values were highly significant after Bonferroni correction. We found large differences in F_{IS} values between demes (Table 3), although each deme did exhibit a heterozygosity deficiency.

Estimates of short and long-term gene flow

Demes differed greatly in short-term estimates of gene flow (from BayesAss+) (Table 4). Estimates ranged from a mean of 0.002 to a mean of 0.06. We found no asymmetric gene flow (nonoverlapping 95 % confidence intervals) for any of the deme pairs. For deme 1, the analysis indicated that the results were close to those being generated by uninformative data (Wilson and Rannala 2003), so we urge caution when BayesAss+ estimates for that deme are considered. Apart from one seemingly high estimate of migration from deme 2 into deme 1, however, the remaining estimates for that deme are not atypical, nor are they biologically un-reasonable. The estimates for the three

Fig. 2 Demes of *P. hubrichti* as determined by Bayesian clustering techniques. Demes are numbered from left to right and represented by unique symbols (1 circles, 2 triangles, 3 stars, 4 squares, 5 diamonds)

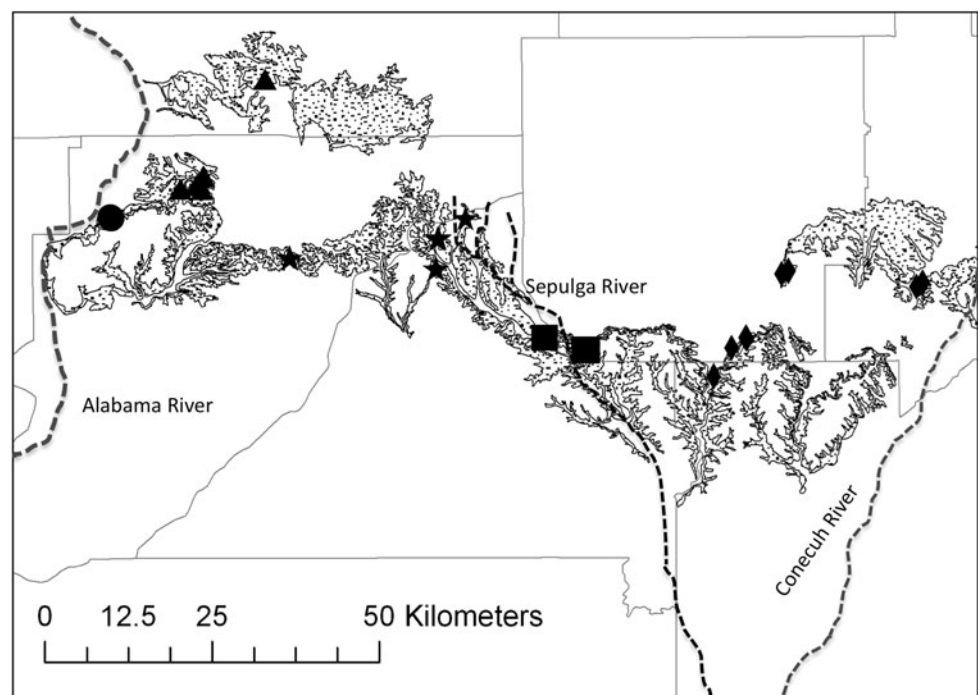


Table 1 Results of the analysis of molecular variance

Source of variation	d.f.	Sum of squares	Variance components	Percentage of variation	<i>P</i> value
Among demes	4	105.717	0.6084 Va	15.58	<0.00001
Within demes	203	669.149	3.2963 Vb	84.42	<0.00001
Total	207	774.865	3.9047		
Fixation index (F_{ST})	0.15582				

P values were determined with 10,000 permutations

Table 2 Pairwise F_{ST} values for *P. hubrichti* demes

Deme	1	2	3	4	5
1	0				
2	0.1423	0			
3	0.1766	0.1313	0		
4	0.1351	0.1372	0.1399	0	
5	0.1675	0.1879	0.1846	0.1301	0

All are statistical significant at the <0.05 level. Significance determined with 10,000 permutations and a sequential Bonferroni correction

Table 3 Summation of allelic information for each of the five demes of *P. hubrichti*

Deme	<i>n</i>	H_O (mean)	H_E (mean)	F_{IS}
1	15	0.5797	0.7342	0.2104
2	17	0.5602	0.6498	0.1378
3	34	0.5669	0.6728	0.1574
4	19	0.5996	0.6723	0.1081
5	20	0.6105	0.6958	0.1226

All F_{IS} values are statistical significance at the <0.05 level

H_O heterozygosity observed, *n* number of individuals per deme, H_E heterozygosity expected, F_{IS} fixation index

runs of BayesAss+ did not vary statistically (*t* tests; $P < 0.05$), indicating that the method produced consistent results pertaining to these data.

Long-term estimates of gene flow from MIGRATE (Beerli 2006) varied less than did short-term estimates (Table 4) but still showed a fair amount of heterogeneity between deme pairs. One deme pair did exhibit a pattern of historically asymmetric migration; deme 3 showed a much higher rate of migration into deme 2 than vice versa. Overall, the migration rates from MIGRATE fall in line with normal expectations (i.e., that neighboring demes have the highest rates of migration), and no demes showed any strong indication of being genetic sinks. Estimates of Θ for all five demes were nearly identical (Table 5), between 0.09 and 0.1. The Wilcoxon test indicated that the BayesAss+ estimates for short-term *m* (migration rate) were significantly lower ($P < 0.001$) than long-term rates

from MIGRATE; most of the values were an order of magnitude lower.

Recent demographic events

We found evidence for recent genetic bottlenecks in up to three of the five demes (Table 6). The IAM model indicated that three demes have experienced bottlenecks in the recent past (Table 6). The sensitivity analysis indicated that two of the demes lost support for a recent bottleneck as more of the SMM model was introduced into the algorithm (Table 6), but because the IAM model is more appropriate given our data (Cornuet and Luikart 1996), we feel that support is strong for at least two of these demes (1 and 4). The support for deme 3 eroded as SMM was added into the model, so it is considered the least likely of the three to have experienced a recent bottleneck.

Landscape analyses

Simple isolation-by-distance analyses (Mantel tests; Mantel 1967) indicated that Euclidian distance and genetic distance were not significantly related for *P. hubrichti* (Table 7), nor were the distance measured along suitable slope and genetic distance, an approach more analogous to how the salamander crawls (rather than how the crow flies). Cost distance and genetic distance were significantly related when we used a model that incorporated slope as a conducting factor in Circuitscape (McRae 2006), compared to the lack of resistance for other habitat, but adding geologic layers into this model weakened the explanatory power (Table 7).

Discussion

Population genetic structuring and diversity

Results from our study revealed that the Red Hills salamander, federally listed as threatened, exhibits a strong pattern of genetic structuring across its entire geographic range. Bayesian clustering identified five distinct and

Table 4 Rates of migration (m) as inferred from a coalescent method (MIGRATE) and from a Bayesian assignment (BayesAss+)

Demes	MIGRATE	95 % confidence interval	BayesAss+	95 % confidence interval
2 → 1	0.0283	0.0108–0.0378	0.0687	0.000004–0.2963
3 → 1	0.0310	0.0108–0.0459	0.0124	0.000003–0.0698
4 → 1	0.0283	0.0081–0.0405	0.0122	0.000003–0.0641
5 → 1	0.0121	0.0000–0.0189	0.0108	0.000004–0.0632
1 → 2	0.0148	0.0000–0.0243	0.0063	0.000001–0.0377
3 → 2	0.0526	0.0297–0.0756	0.0075	0.000001–0.0450
4 → 2	0.0256	0.0054–0.0350	0.0061	0.000001–0.0358
5 → 2	0.0256	0.0081–0.0351	0.0059	0.000001–0.0376
1 → 3	0.0121	0.0000–0.0189	0.0029	0.000004–0.0150
2 → 3	0.0094	0.0000–0.0162	0.0059	0.000051–0.0231
4 → 3	0.0175	0.0000–0.0243	0.0028	0.000002–0.0149
5 → 3	0.0121	0.0000–0.0189	0.0025	0.000001–0.0147
1 → 4	0.0256	0.0054–0.0351	0.0053	0.000001–0.0298
2 → 4	0.0202	0.0027–0.0270	0.0062	0.000001–0.0339
3 → 4	0.0175	0.0000–0.0270	0.0098	0.000003–0.0475
5 → 4	0.0337	0.0162–0.0432	0.0054	0.000001–0.0305
1 → 5	0.0148	0.0000–0.0243	0.0056	0.000001–0.0324
2 → 5	0.0175	0.0000–0.0270	0.0054	0.000002–0.0293
3 → 5	0.0202	0.0000–0.0297	0.0134	0.000105–0.0489
4 → 5	0.0202	0.0027–0.0297	0.0061	0.000003–0.0350

All rates are represented as m

Table 5 Estimates of Θ and effective population size (N_e) for all demes of *P. hubrichti*

Deme	Θ ($4N_e\mu$)	95 % confidence interval	N_e
1	0.0972	0.0885–0.1	45.0231
2	0.0977	0.0905–0.1	45.2546
3	0.0987	0.0950–0.1	45.7175
4	0.0982	0.0930–0.1	45.4861
5	0.0982	0.0925–0.1	45.4861

N_e was estimated from Θ by means of a well-accepted vertebrate mutation rate of 5.4×10^{-4} (Goldstein et al. 1995; Howes et al. 2009)

well-supported demes (Fig. 2). These groupings were highly supported by an abundance of evidence, including high F_{ST} values between demes, high support from likelihood-assignment tests, and AMOVA results (Table 2). Such a high degree of spatial structuring across the range of a species with the characteristics of *P. hubrichti* is not surprising, but our previous knowledge of the species' distribution and genetic history (Schwaner and Mount 1970; McKnight et al. 1991) did not lead us to expect the geographic nature of the deme structuring. For example, mitochondrial data have suggested only two main lineages within the species (McKnight et al. 1991). Previous surveys have shown that the three geologic formations to which *P. hubrichti* is closely tied are all “claystone,” which

retains moisture and sustains burrow shape (Schwaner and Mount 1970; French and Mount 1978; Dodd 1991; Apodaca 2010). Nonetheless, our results demonstrate that the absence of these geologic formations does not necessarily preclude gene flow (Fig. 2; Tables 2, 4). In addition, previous research with mtDNA has suggested that large rivers impede gene flow (McKnight et al. 1991), but our results provide little support for this hypothesis (Fig. 2; Tables 2, 4). In fact, demes seem to be maintained by the continuous presence of suitable slope, which often accompanies river drainages.

Estimates of short- and long-term gene flow

Estimates of long-term rates of migration were fairly low (Table 4) but are similar to values found for other low-vagility amphibians (e.g. by Wang 2009; Blouin et al. 2010), so even before human disturbance, *P. hubrichti* dispersal events may have been uncommon and probably occurred by a stepping-stone process. Estimates of short-term migration rates were significantly lower than those of long-term rates, however (Wilcoxon matched-pairs test; $P < 0.001$). By comparing these rates (Table 4), we can infer that migration rates in the short-term past have declined. Given that coalescent-based methods estimate average population parameters over roughly the last $4N_e$ generations (Wilson and Rannala 2003) and that all demes in the study had an N_e 's above 40, we presume that

estimates from MIGRATE corresponded to gene flow before major human habitat destruction in the region (about the late 1800s). It should be noted that these estimates are dependent on adequately sampling a population's true genetic diversity. Low sample sizes could certainly be affecting our migration and effective population size estimates, however, given that estimates of migration across the species range show a similar pattern we feel confident that our results are not strongly biased by low sample sizes. Furthermore, other analyses corroborate strongly the conclusion that gene flow has been reduced since wide-spread habitat destruction by humans throughout the range of *P. hubrichti*. For example, a previous fragmentation analysis indicated that *P. hubrichti* had lost between 69.5 and 86.1 % of its original habitat (Apodaca 2010). Losing such a large amount of habitat could hardly fail to affect a species' migration rates, especially in a naturally fragmented species with low dispersal ability. However, the most compelling evidence of such effects would come from the incorporation of fragmentation data into a landscape genetics study. While this is currently difficult, the rate at which such data is becoming available will surely benefit such studies in the future.

The effects of fragmentation can also be seen through an examination of recent demographic events, which reveal that multiple demes have undergone a recent bottleneck (Table 6). Our estimate of the number of such demes is probably low, because the two demes that were not identified as having undergone a recent bottleneck (demes 2 and 5; Table 6) break one of the key assumptions of bottleneck analyses, the presences of subpopulation structure (Busch et al. 2007; Marshall et al. 2009). Estimates of very small N_e for all demes (Table 5) support the evidence for recent bottlenecks, but low N_e could also result from factors such as low sample sizes or from the species natural history (i.e. a strong reproductive skew or uneven sex ratios) (Beebe 2005), both of which are not known to occur in *P. hubrichti*. Effective population sizes in this range are rare in natural populations, and the other amphibians with similar N_e are either threatened species or found in populations that have experienced heavy fragmentation (see, e.g., Rowe and Beebe 2004; Funk et al. 1999; Wang 2009). Still, some amphibian species that do not appear to be facing an imminent threat exhibit similar N_e values (see, e.g., Funk et al. 1999; Jehle et al. 2001), probably because of poor dispersal ability or the high site fidelity that amphibians display.

Conservation implications

Our study has several important implications for the recovery and management of *P. hubrichti*. Protection of continuous slope habitat is seemingly an important element

Table 6 Results of bottleneck analysis for all demes of *P. hubrichti*

Deme	Model	Percentage of stepwise mutation model	Probability of heterozygosity deficit	Probability of heterozygosity excess
1	IAM	0	0.9877	0.0161*
2	IAM	0	0.5771	0.4609
3	IAM	0	0.9877	0.0161*
4	IAM	0	0.9975	0.0034*
5	IAM	0	0.9033	0.1162
1	TPM	10	0.9580	0.0527*
2	TPM	10	0.2783	0.7539
3	TPM	10	0.8837	0.1377
4	TPM	10	0.9965	0.0048*
5	TPM	10	0.7841	0.2460
1	TPM	20	0.9580	0.0527*
2	TPM	20	0.2783	0.7539
3	TPM	20	0.8388	0.1875
4	TPM	20	0.9965	0.0048*
5	TPM	20	0.7841	0.2460
1	TPM	30	0.9472	0.0654
2	TPM	30	0.2783	0.7539
3	TPM	30	0.7841	0.2460
4	TPM	30	0.9951	0.0068*
5	TPM	30	0.7539	0.2783
1	TPM	40	0.9472	0.0654
2	TPM	40	0.2158	0.8125
3	TPM	40	0.8623	0.1611
4	TPM	40	0.9877	0.0161*
5	TPM	40	0.7216	0.3125

All values are presented as *P*-values. Statistically significant values are marked with asterisks

IAM infinite-alleles model, *TPM* two-phase model

Table 7 Results of Mantel tests for correlations between geographic/model distance and genetic distance

Model	Included variables	r^2
Euclidian distance	None	0.306
Slope distance	None	0.287
Slope conductance	Slope	0.321*
Slope and geology	Slope, geologic formations	0.281

Statistically significant values with 10,000 permutations are marked with asterisks

in any recovery plan for this species given that this species inhabits and disperses primarily along slope habitat; as demonstrated by our landscape genetics analysis (Table 7). It should be noted that we were not able to assess the amount of habitat destruction directly along slope habitat. Quantifying the effects that transforming slope habitat has on *P. hubrichti* should be a focus of future research.

However, we provide evidence for the negative effects of range wide habitat fragmentation by demonstration a dramatic decline in short-term migration rates and the presence of bottlenecks. Clearly, the long-term survival of this species depends on restoring gene flow between populations. This is likely dependent on the restoration of intervening habitat. Such restoration would probably help restore historical levels of gene flow and alleviate the pressures of inbreeding depression that can occur in small populations (Frankham et al. 2002). Despite the high percentage of current habitat that has been lost or altered (Apodaca 2010), we cannot estimate how much of the habitat has been negatively effected by forestry practices such as clear-cutting to the edge of the hardwood canopy or chemical preparation of surrounding areas. We therefore urge the adoption of the habitat guidelines laid out by Dodd (1991) and Apodaca (2010). These guidelines include, but are not limited to, avoiding clear cutting and mechanical and/or chemical preparation on or near inhabited slopes. Additionally, efforts should be made to maintain the natural hardwood assemblage found on inhabited slopes.

The future status of *P. hubrichti* depends strongly on the establishment of protected areas. During nearly the entire 35 years that *P. hubrichti* has been listed by the U.S. Fish and Wildlife Service as a threatened species, the only protection afforded them has been in the form of habitat conservation plans (HCPs), generally with large timber companies. Although HCPs are a viable tool for the conservation of a species, they do not always incorporate sufficient scientific data to ensure species persistence (Harding et al. 2001). In fact, most current *P. hubrichti* HCPs allow for selective harvest on any habitat with slope less than 28°. This suboptimal habitat may not harbor the highest numbers of individuals, but it is probably vital to gene flow between populations.

Conclusions

Our study demonstrates the importance of considering history during analysis of the genetic patterns of a species and the inherent conservation implications of such results. Recent advances in landscape genetics (Manel et al. 2003) have demonstrated that incorporating multiple sources of data (such as landscape resistance) into a population genetics framework is a useful endeavor (see, e.g., Storfer et al. 2007; Lee-Yaw et al. 2009; Wang 2009; Wang and Summers 2010). We have demonstrated that, in addition to consideration of landscape, weighing the effects of life-history characteristics and the history of habitat alteration on genetic structuring and gene flow between populations is valuable. We have shown that *P. hubrichti* naturally exhibits a low migration rate and therefore strong

population structuring. Using these baseline data, we have demonstrated that short-term migration rates between populations are even lower than would be expected from the species' low vagility. Clearly, habitat modification has had a negative effect on the species; it does not merely suffer from the life-history characteristics of a naturally fragmented species.

Acknowledgments We thank R. Crawford, J. Mitchell, C. Tucker, and A. Waggoner for assistance with laboratory work. C. Thawley, H. Cunningham, and W. Smith provided assistance in the field and helpful comments. M. Wines, W. Callans, and S. Hoss assisted with field work. We also thank M. Gunzburger, K. Dodd, J. Travis, and J. Homyack for other assistance and advice. This project was funded by a State Wildlife Grant through the State of Alabama Division of Wildlife and Freshwater Fisheries. All tissue collection was performed under Federal Fish and Wildlife permit # TE136961-0 and State of Alabama permit # AWFF-SPS-08-02. Several timber companies allowed access to their property for this project, including Weyerhaeuser, Wilmon Timber, Resource Management, and Lawley Timber Company. Three anonymous reviewers also provided helpful comments on this version of the manuscript and A. Thistle provided editorial services.

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